

Treatment Facilities and Drug Mortality Across Opioid Eras: A Replication and Extension

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September 2026

Abstract

An influential estimate from the early opioid epidemic indicates that additional substance-use treatment facilities reduced county drug mortality. I replicate that estimate and test whether its county-level relationship changes after the original 1999–2008 period. The County Business Patterns estimate reproduces almost exactly: one additional lagged establishment is associated with 0.035 fewer drug-induced deaths per 100,000 residents. A pooled coefficient-equality test shows that the slope changes by 0.077 after 2008 ($p = 0.003$), from -0.035 to an individually imprecise 0.042. A separately constructed federal treatment-directory exposure produces a nearly identical early coefficient and also rejects stability. Continuous-treatment estimators nevertheless fail every joint pre-change placebo test, including in the original period. Matching methadone-providing facilities to opioid-involved deaths does not recover a protective relationship either. These results do not imply that treatment became ineffective or harmful. They show that the early county-level association changes in later opioid eras, while mortality dynamics preceding facility changes prevent a strong causal interpretation.

1 Introduction

Drug-overdose mortality continued to rise as the United States expanded its substance-use treatment infrastructure. The national overdose rate increased from 5.95 deaths per 100,000 residents in 1999 to 21.43 in 2019. Its composition changed even more dramatically. Mortality involving synthetic opioids other than methadone rose from 0.26 to 11.01 deaths per 100,000, and deaths involving cocaine and psychostimulants became increasingly common. Over roughly the same period, the number of outpatient and residential treatment facilities grew. These aggregate trends do not establish that treatment is ineffective: mortality could have risen faster without that expansion. They instead motivate a narrower question. Does the protective county-level relationship documented during the early opioid epidemic persist as the drug market and treatment system evolve?

This paper revisits the relationship between local treatment-facility availability and drug mortality established by Swensen (2015). Swensen uses changes in county treatment-establishment counts from 1999 through 2008 and finds that additional facilities reduce drug-induced mortality. That result remains relevant to policy. State and federal responses to the opioid epidemic continue to devote substantial resources to expanding treatment supply, and later studies use related access variation to examine crime, emergency-department visits, and syringe-exchange programs (Bondurant et al., 2018; Corredor-Waldron and Currie, 2022; Packham, 2022). Yet the object measured by one additional facility may have changed. Facilities differ in capacity and services, medication treatment increasingly occurs outside traditional specialty providers, and the illicit market shifted from prescription opioids toward heroin, fentanyl, and polysubstance use (Maclean et al., 2022; Ciccarone, 2019, 2021).

The question is deliberately different from whether substance-use treatment “works.” Retention in methadone or buprenorphine treatment is associated with substantially lower all-cause and overdose mortality (Sordo et al., 2017; Larochelle et al., 2018). A county facility count instead asks whether one more establishment expands effective access enough to change population mortality. The two relationships can diverge. A new facility may bring untreated people into care, reduce travel costs, or add scarce treatment slots. It may also redistribute existing patients, enter with little capacity, or respond to a local increase in substance-use severity. Clinical efficacy can therefore

remain high even when the marginal county-level relationship becomes small or confounded.

I first reproduce Swensen’s 1999–2008 specification using County Business Patterns (CBP) counts of outpatient and residential treatment establishments, then extend the same design through 2016 and 2019. Detailed CBP county-industry coverage deteriorates after 2016, so I construct an independent county panel from annual SAMHSA/N-SSATS treatment directories. I standardize facility listings and assign counties from street addresses, using ZIP-derived geography only when a defensible street match is unavailable. Rather than attach directory observations to the end of CBP, I compare the two sources on identical county-year support and use the directory consistently throughout its extension sample. I also reconsider the comparison using methods designed for continuous treatments that can rise and fall and match the directory’s methadone-provision measure to opioid-involved mortality.

The original estimate replicates, but the relationship changes after its sample period. One additional lagged CBP establishment in 1999–2008 is associated with 0.0352 fewer drug-induced deaths per 100,000 residents, nearly identical to Swensen’s estimate. On the common CBP–directory support through 2016, the CBP slope changes from -0.0359 before 2009 to 0.0418 afterward. The difference is 0.0777 ($p = 0.0025$). The directory reproduces the early CBP relationship on that same support and also rejects coefficient stability through 2016 and 2019. The later positive coefficients are individually imprecise, so the evidence establishes a change in the estimated association rather than a harmful effect of additional facilities.

Recent difference-in-differences methods sharpen the design problem but do not overturn that conclusion. Facility counts are continuous and nonabsorbing: counties can gain several facilities, lose them, or reverse prior changes. Conventional two-way fixed-effects estimates can consequently combine comparisons that receive different weights when effects vary across treatment histories. I estimate normalized dynamic effects using de Chaisemartin and D’Haultfoeuille (2022) and interpret them in light of Callaway et al. (2022). In several samples the resulting estimates are negative. But joint tests of the three pre-change placebo coefficients reject in every specification, including the original 1999–2008 CBP window. Counties were already experiencing differential changes in mortality before their facility counts changed. The modern estimates are therefore informative diagnostics, not a preferred causal replacement for the original coefficient.

An outcome-matched extension reaches the same limit. Beginning in 2002, the directory reports

whether a facility provides methadone, allowing the lagged service measure to be followed against mortality from 2003 through 2019. Holding the total facility count fixed, an additional methadone-providing facility is associated with 0.098 additional opioid-involved overdose deaths per 100,000 residents, with a standard error of 0.098. The estimate is imprecise and does not show that methadone raises mortality. Its sign is instead consistent with methadone services responding to local opioid need. The corresponding association for overdoses without recorded opioid involvement is negative, while the coefficient for all drug overdoses is essentially zero. Matching the service and outcome more closely does not recover a protective county-level estimate.

The paper separates three questions that an expanding-window replication can otherwise confound. First, does the published coefficient reproduce? It does, in CBP and in an independently constructed directory measure. Second, does that relationship transport beyond the original period? Formal pooled tests reject stability in both sources. Like Hansen et al. (2017), this comparison holds the inherited design fixed long enough to test its persistence. Third, can modern weighting or closer service–outcome matching supply a credible causal interpretation? Rejected pre-change tests and the methadone results indicate that they cannot. The contribution is therefore not a new estimate of treatment effectiveness. It is evidence that replication, transportability, and identification reach different conclusions in this setting.

2 From Clinical Efficacy to County Treatment Infrastructure

The treatment measure in this paper captures local infrastructure rather than treatment received by a particular patient. CBP counts establishments in outpatient and residential substance-use industries, while the federal directories count listed treatment facilities. Both measures combine providers that differ in capacity, quality, patient population, and services. Neither records whether the marginal facility increases the number of patients treated. An opening can improve access by reducing travel and search costs, expanding available slots, or introducing a service that was previously unavailable. It can have little effect on realized access if it reallocates patients across providers or enters with limited capacity.

This distinction separates the paper’s estimand from the clinical efficacy of medications for opioid use disorder (MOUD). Methadone and buprenorphine are associated with substantially lower

mortality while patients remain in treatment (Sordo et al., 2017; Larochelle et al., 2018). I do not estimate those patient-level effects. The replication treatment combines specialty facilities serving opioid, alcohol, and other substance-use disorders, and MOUD can also be delivered outside those facilities. The county coefficient therefore reflects several margins at once: provider entry, capacity, service composition, treatment initiation, retention, and potentially the local demand that induced entry.

The directory allows me to narrow this mismatch. It identifies methadone provision beginning in survey year 2002. Restricted multiple-cause mortality records identify drug-overdose deaths involving opioids and a mutually exclusive category without recorded opioid involvement. The non-opioid outcome is useful because it changes much less over the sample, but it is not a placebo. Specialty facilities treat multiple substance-use disorders, deaths may involve several drugs, and contributing substances may be incompletely recorded. Certified opioid treatment program (OTP) status would define the provider more narrowly, but the reliable county-directory code begins only in 2015. The earlier official survey file contains an OTP field without county geography and cannot be linked to the directory records. A county OTP analysis would consequently contain only four outcome years and is not suitable as the principal extension.

The relationship between facility entry and mortality may also have changed with the drug supply. The early opioid epidemic was dominated first by prescription opioids and then by heroin. Beginning in the mid-2010s, illicit fentanyl drove a rapidly increasing share of deaths, followed by growing opioid-stimulant involvement (Ciccarone, 2019, 2021). A more lethal and uncertain drug supply can increase the value of effective treatment, but it can also raise mortality before treatment begins and make treatment continuity more consequential. These changes motivate testing whether the earlier relationship persists; they do not by themselves identify why it changes.

Treatment financing and delivery evolved at the same time. Medicaid expansions increased treatment admissions, entry by selected medication-treatment providers, and Medicaid acceptance among existing providers (Meinhofer and Witman, 2018). Yet Grooms and Ortega (2019) finds increased Medicaid-covered admissions without a detectable increase in the total number of treatment facilities, while Barrette et al. (2023) shows that treatment composition can shift when insurance rules and prescribing access change. Utilization can therefore expand through payer acceptance, capacity, and service mix without proportional facility growth. Conversely, treatment supply can

respond to worsening local conditions. DiNardi (2025) asks how OxyContin’s reformulation shifted treatment supply and finds faster outpatient-facility growth in more exposed states. I instead ask whether county facility counts retain their mortality relationship. Synthetic-opioid mortality shocks also increased buprenorphine prescribing where prescribing capacity already existed (Siegal et al., 2026). These responses make pre-entry mortality dynamics a central empirical concern.

I end the analysis in 2019 because the COVID-19 pandemic created a distinct mortality and treatment-delivery regime. Overdose deaths rose abruptly, while social isolation, economic disruption, and changes in substance use altered overdose risk (Friedman and Akre, 2021; Roberts et al., 2021). Federal emergency rules simultaneously expanded telehealth access to buprenorphine and permitted longer take-home supplies of methadone; national methadone supply fell in early 2020 while buprenorphine supply continued to grow (Chen et al., 2022). The pandemic also generated geographically heterogeneous policy responses that complicate difference-in-differences comparisons (Goodman-Bacon and Marcus, 2020). Ending in 2019 preserves a pre-pandemic extension in which a change in facility counts has a more comparable institutional meaning.

3 Measuring and Estimating the County Facility Relationship

3.1 County mortality and two independent facility counts

The unit of observation is a county-year. The replication outcome is the number of drug-induced deaths per 100,000 residents, constructed from restricted National Center for Health Statistics mortality records and county population. This is the outcome used in Swensen’s preferred specification. The analysis also includes the original demographic and economic controls: unemployment, births, per-capita income, racial and sex composition, and six age shares. The restricted mortality records and derived county panel cannot be distributed publicly; the replication materials instead document the construction and provide links for obtaining public, suppressed mortality aggregates.

The replication exposure is the one-year-lagged count of outpatient and residential substance-use treatment establishments in CBP. The extension exposure is the one-year-lagged count of facilities in SAMHSA/N-SSATS treatment directories parsed for 1998–2019. CBP establishments and directory facilities are related but different reporting units. I estimate their coefficients separately and never append or splice the series.

The directory contains 238,148 facility-year records, of which 238,074 retain a complete street address, city, state, and ZIP code. After address normalization, the 63,045 unique complete addresses yield 51,809 high-confidence Census street matches; 823 are flagged for review and 10,413 are terminal unmatched. The primary county assignment uses the street match when available and a documented ZIP-derived county otherwise. At least 89.45 percent of facility rows in every year from 2015 through 2019 receive a high-confidence street county. Among street-matched rows with an original ZIP county, the street assignment changes the county for approximately 1.6–1.7 percent. I retain street-only and original-ZIP measures as geography checks.

Survey year 2001 is absent from the parsed-directory sequence. I preserve this gap rather than interpolate a count; because treatment is lagged, directory specifications omit mortality year 2002. The exact CBP replication covers 1999–2008. The early matched comparison uses county-years observed in both sources over that period. The longer matched comparison continues through 2016, the last year of broad CBP county-industry coverage, and the directory extension continues through 2019. Directory models use the 2,524-county universe recovered from the replication sample.

The service extension counts facilities reporting methadone provision. That field begins in 2002, producing a 2003–2019 mortality window after the lag. Buprenorphine availability begins later and is retained as a secondary measure rather than a main exhibit. The opioid-involved outcome requires an underlying drug-overdose death and any contributing ICD-10 code T40.0–T40.4 or T40.6. The comparison outcome contains overdose deaths with no recorded opioid code. These mutually exclusive categories reconstruct all drug-overdose deaths in each county-year. Nationally, non-opioid overdose mortality rises from 3.10 to 6.32 deaths per 100,000 between 1999 and 2019, compared with an increase from 2.85 to 15.11 for opioid-involved mortality.

3.2 Window comparisons and a pooled stability test

Figure 1 evaluates whether the directory provides a credible independent measure before the CBP break. The annual cross-sectional correlation between county CBP and directory levels ranges from 0.896 to 0.946. In contrast, the correlation between annual within-county changes ranges from -0.198 to 0.181 . National totals also grow at different rates: relative to 1998, the published CBP count reaches 1.53 in 2016, while the directory count reaches 1.19. The agreement in levels shows that the sources measure the same broad treatment market. Their weak agreement in changes

makes it especially important to estimate them separately on common support.

For each source and sample, I estimate

$$y_{ct} = \beta F_{c,t-1} + X'_{ct}\gamma + \alpha_c + \lambda_{s(c),t} + \varepsilon_{ct}, \quad (1)$$

where y_{ct} is drug-induced mortality per 100,000 residents, $F_{c,t-1}$ is the lagged CBP or directory count, X_{ct} contains the Swensen controls, α_c denotes county fixed effects, and $\lambda_{s(c),t}$ denotes state-by-year fixed effects. Regressions are weighted by county population and standard errors are clustered by county. In the level specification, β is the mortality-rate change associated with one additional lagged establishment or facility under the maintained design.

The matched-source comparisons hold the outcome, controls, fixed effects, weights, clustering, counties, and years fixed. Table 1 places the original replication coefficient beside the early- and later-period coefficients from the extension samples. It then reports their difference directly. This presentation distinguishes replication of the published estimate from the separate question of whether the relationship persists after the original sample period.

I test coefficient stability in a pooled Chow specification:

$$y_{ct} = \beta_E F_{c,t-1} + \delta(F_{c,t-1} \times P_t) + X'_{ct}\gamma_E + (X_{ct} \times P_t)' \gamma_\Delta + \alpha_{c,e(t)} + \lambda_{s(c),t} + \varepsilon_{ct}, \quad (2)$$

where P_t equals one for outcome years beginning in 2009 and $\alpha_{c,e(t)}$ denotes county-by-era fixed effects. The 2008 boundary is inherited from Swensen's published sample rather than selected from the extended outcome path. The early slope is β_E , the later slope is $\beta_E + \delta$, and the equality test is $H_0 : \delta = 0$. Allowing county intercepts and control slopes to differ across eras makes the two slopes exactly equal to estimates from separate early- and later-period regressions, while estimation in one model preserves their county-level covariance. I estimate Equation (2) using the exact CBP support through 2016, matched CBP and directory support through 2016, and the directory through 2019. Population weights and county clustering remain unchanged. The expanding-window coefficient in Equation (1) and the later-period coefficient in Equation (2) therefore answer different questions. The online appendix reports more restrictive interaction models and the weighted-SD and $\log(1+F)$ transformations.

3.3 Continuous-treatment and service-matched diagnostics

I complement Equation (1) with the continuous, nonabsorbing dynamic difference-in-differences estimator implemented by `did_multiplegt_dyn`. The estimator permits facility counts to increase, decrease, and reverse; it reports normalized effects following a count change and placebo effects preceding that change. Models use population weights, state-specific comparison strata, and county-clustered standard errors. To avoid bridging the missing directory year, matched robust samples use consecutive years beginning in 2003. The twelve regression controls are not included because combining them with state-specific comparison cells produces singular internal regressions. The resulting estimates are not coefficient-for-coefficient replacements for β . Their central contribution is a direct test of whether mortality was already changing before the observed facility-count change.

For the methadone analysis, I estimate versions of Equation (1) that include the total directory count and the methadone-providing count together. The methadone coefficient therefore captures the incremental association of one additional methadone provider, holding the total number of facilities fixed. I estimate identical 2003–2019 models for opioid-involved mortality, overdose mortality without recorded opioid involvement, and their sum.

4 The Early Association Changes, but Causal Interpretation Remains Unwarranted

4.1 The early estimate replicates and the pooled test rejects stability

Table 1 makes the replication and extension comparison directly. The first row reports the exact 1999–2008 CBP coefficient of -0.03520 (standard error 0.00637) across 25,214 county-years. It reproduces the previously validated Swensen coefficient to machine precision. The next row uses the common CBP–directory support through 2016. The early CBP coefficient is -0.03588 (standard error 0.00701), and the post-2008 coefficient is 0.04181 (standard error 0.02436). The difference is 0.07769 , with a standard error of 0.02569 and a 95 percent confidence interval of $[0.02731, 0.12808]$; the equality-test p -value is 0.0025 .

The final two rows show the same pattern using the separately measured directory exposure. On matched support through 2016, its slope changes from -0.03776 to 0.01256 . The difference

is 0.05032, with a 95 percent confidence interval of $[0.02925, 0.07139]$ and $p < 0.0001$. Through 2019, the later directory slope is 0.02272 and the difference from the early coefficient is 0.06048, with a 95 percent confidence interval of $[0.03482, 0.08615]$ and $p < 0.0001$. The later slopes are not individually distinguishable from zero at the 5 percent level. The agreement across exposure measures corroborates the change but is not independent outcome evidence: both models use the same mortality panel. The result is therefore a statistically significant change away from the early negative association, not evidence of a positive later treatment effect.

Alternative county assignments do not explain the 2019 result. The street-only coefficient is -0.00387 , and the original-ZIP coefficient is -0.00323 , compared with -0.00308 for the street-plus-fallback measure. The matched 1999–2016 estimates are similarly close across geography rules. ZIP-derived geography can assign a facility to the wrong county when a ZIP crosses county lines, but replacing it with address-based street matches leaves the substantive result unchanged.

The online appendix shows that a more restrictive pooled interaction with one county intercept across both eras does not recover the original-period coefficient and produces a different comparison. I use the fully interacted Chow model because it estimates the difference between the two period-specific coefficients that motivate the exercise. This sensitivity is not innocuous: it reinforces that the result concerns transportability of an observational specification rather than a structural treatment effect. The appendix also shows that the early negative relationship is strongest in the linear level-count specification inherited from Swensen and is not a scale-invariant elasticity.

4.2 Modern estimates do not pass the design’s placebo tests

Figure 2 reports the continuous-treatment results. In the exact CBP window, the normalized average cumulative effect is -0.264 (standard error 0.092). In 2003–2016, the CBP estimate is 0.068 (standard error 0.109), while the directory estimate is -0.399 (standard error 0.159). Extending the directory through 2019 yields -0.447 (standard error 0.156). Because these estimates normalize effects by treatment dose and weight treatment histories differently, their magnitudes should not be compared directly with the raw TWFE coefficient.

Three pre-change placebo coefficients determine how much causal weight these results can bear. Using county-clustered inference, the joint placebo p -value is 0.0002 in the original CBP window, below 0.0001 for CBP in 2003–2016, 0.0059 for the directory in 2003–2016, and 0.0295 for the

directory through 2019. Every specification rejects at the 5 percent level. The failure in the original window is especially informative. It shows that differential pre-trends are not created only by the fentanyl era, the longer sample, or the directory measure. Instead, mortality changes before observed facility entry and exit under both measures. Modern weighting changes the estimand, but it cannot make those facility changes conditionally exogenous.

The close early CBP and directory coefficients show that the published association is reproducible and not unique to one administrative source. The placebo paths show that reproducibility is not equivalent to identification. I therefore retain the TWFE sequence as the most direct test of whether the original empirical relationship transports to later years and use the continuous-treatment estimates to diagnose, rather than claim to solve, the underlying design problem.

4.3 Methadone provision is not negatively associated with opioid mortality

Figure 3 reports the outcome-specific methadone estimates. Holding the total directory count fixed, one additional lagged methadone-providing facility is associated with 0.0981 additional opioid-involved overdose deaths per 100,000 residents, with a standard error of 0.0975. The 95 percent confidence interval, $[-0.0930, 0.2893]$, includes zero and economically meaningful effects in both directions. The corresponding coefficient for overdose deaths without recorded opioid involvement is -0.0925 (standard error 0.0384). Because the outcomes partition all fatal drug overdoses, their coefficients sum to an all-overdose estimate of 0.0057 (standard error 0.0871).

The contrast is not evidence that methadone increases opioid mortality or prevents non-opioid overdose. In particular, the negative non-opioid coefficient is not a valid placebo effect because facilities treat multiple disorders and drug involvement is imperfectly recorded on death certificates. Methadone provision is not assigned randomly, and facilities may enter or add services in response to local opioid demand. The positive opioid-specific coefficient alongside the negative non-opioid coefficient is difficult to reconcile with a causal treatment interpretation but easy to reconcile with demand-responsive service placement. More closely matching the facility service to the relevant mortality outcome does not produce the protective relationship one would expect if provider additions were quasi-random. The four-year county OTP panel is too short to provide a more credible alternative.

4.4 The extension spans a different overdose environment

Figure 4 provides the context for the attenuating coefficient. The national all-drug overdose rate rises from 5.95 deaths per 100,000 in 1999 to 12.00 in 2008 and 21.43 in 2019. Mortality involving synthetic opioids other than methadone rises from 0.26 to 0.76 and then 11.01 over the same years. By 2019, mortality involving cocaine and psychostimulants reaches 4.81 and 4.94 deaths per 100,000, respectively. Heroin mortality rises through 2016 and then declines modestly. Drug-specific series overlap because a death can involve more than one substance.

These trends do not identify the mechanism behind the changing facility coefficient. They establish that the extension evaluates the original design against an outcome whose level and drug composition differ sharply from 1999–2008. They also clarify the measurement problem. By 2019, fatal overdoses are increasingly concentrated in synthetic opioids and polysubstance use, while the baseline exposure continues to assign the same unit value to heterogeneous treatment establishments. The methadone analysis narrows that mismatch but cannot remove endogenous service placement.

5 Implications for Interpreting Facility-Count Estimates

The early relationship between treatment-facility counts and drug mortality is reproducible but does not transport unchanged to later years. I recover Swensen’s original estimate using CBP and obtain a nearly identical coefficient from an independently constructed treatment directory on common support. Pooled slope-equality tests reject equality between the early and post-2008 slopes in both sources. The exact CBP difference is 0.0770, and the directory difference is 0.0503 through 2016 and 0.0605 through 2019. The later point estimates are positive but individually imprecise. Thus, the evidence establishes instability in the county-level association, not a harmful effect of additional treatment facilities.

Modern continuous-treatment estimates do not restore a credible causal interpretation. Their placebo tests reject in every sample, including the original period, indicating that mortality was already changing before counties gained or lost facilities. The methadone extension adds a different diagnostic: matching a treatment service to opioid-involved deaths does not produce the expected protective association, and the pattern is consistent with services responding to local need.

These findings do not show that treatment or MOUD ceased to be effective. They show that a reproducible facility-count coefficient can change across periods while remaining difficult to interpret causally in either period. For policy, the distinction matters. Expanding effective treatment may operate through capacity, payer acceptance, medication provision, initiation, and retention without producing one-for-one changes in establishment counts. Future population-level evaluations therefore need either plausibly exogenous changes in treatment supply or data that measure those access margins directly. The original estimate remains an important benchmark; the extension identifies where its empirical relationship stops transporting and why that change should not be read as a clinical treatment effect.

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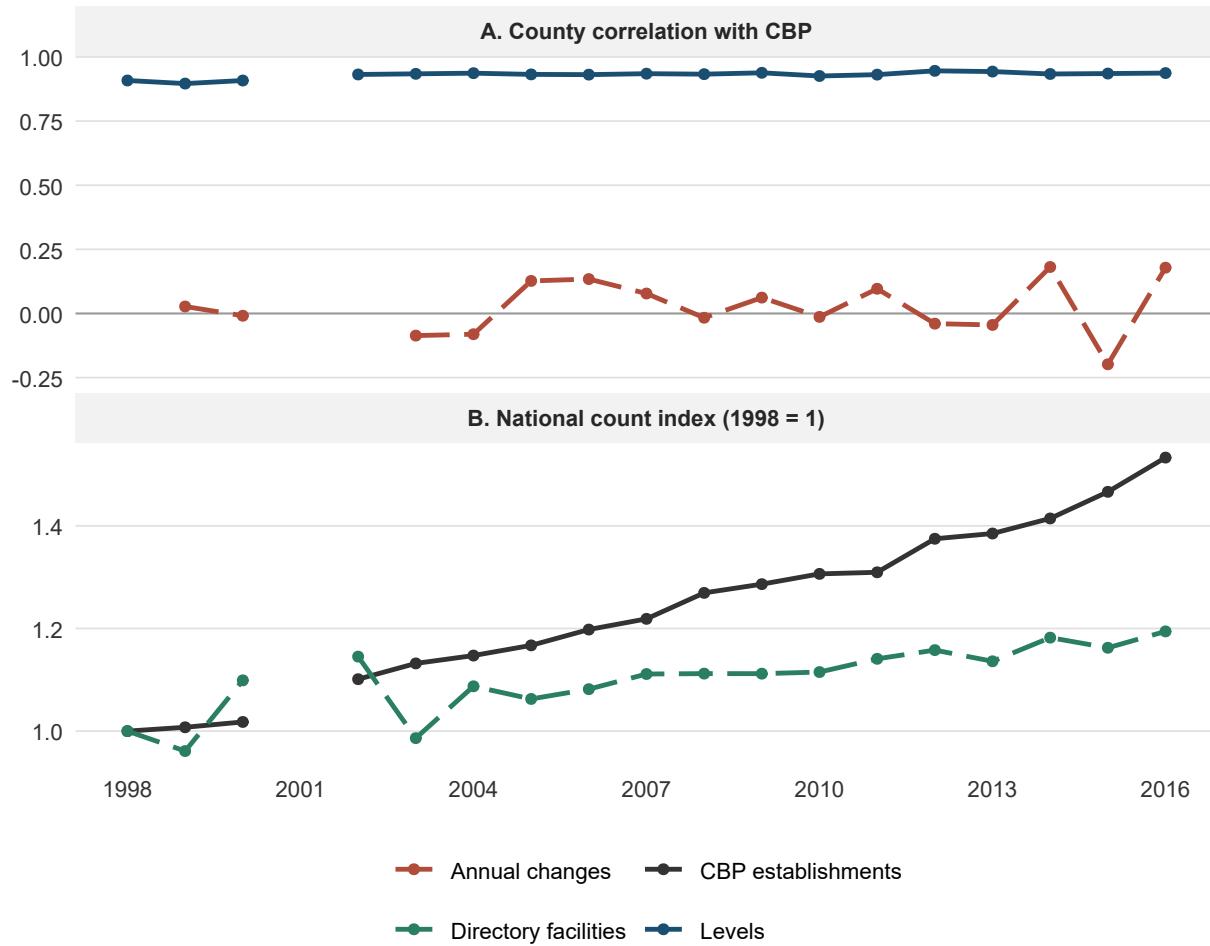
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Table 1. Replication and Extension of the Facility–Mortality Relationship

Sample and source	Early-period coefficient	Later-period coefficient	Later minus early	p -value for equality
CBP, original sample (1999–2008)	-0.0352 (0.0064)	—	—	—
CBP, matched through 2016	-0.0359 (0.0070)	0.0418 (0.0244)	0.0777 (0.0257)	0.0025
Directory, matched through 2016	-0.0378 (0.0062)	0.0126 (0.0086)	0.0503 (0.0107)	< 0.001
Directory, extended through 2019	-0.0378 (0.0062)	0.0227 (0.0123)	0.0605 (0.0131)	< 0.001

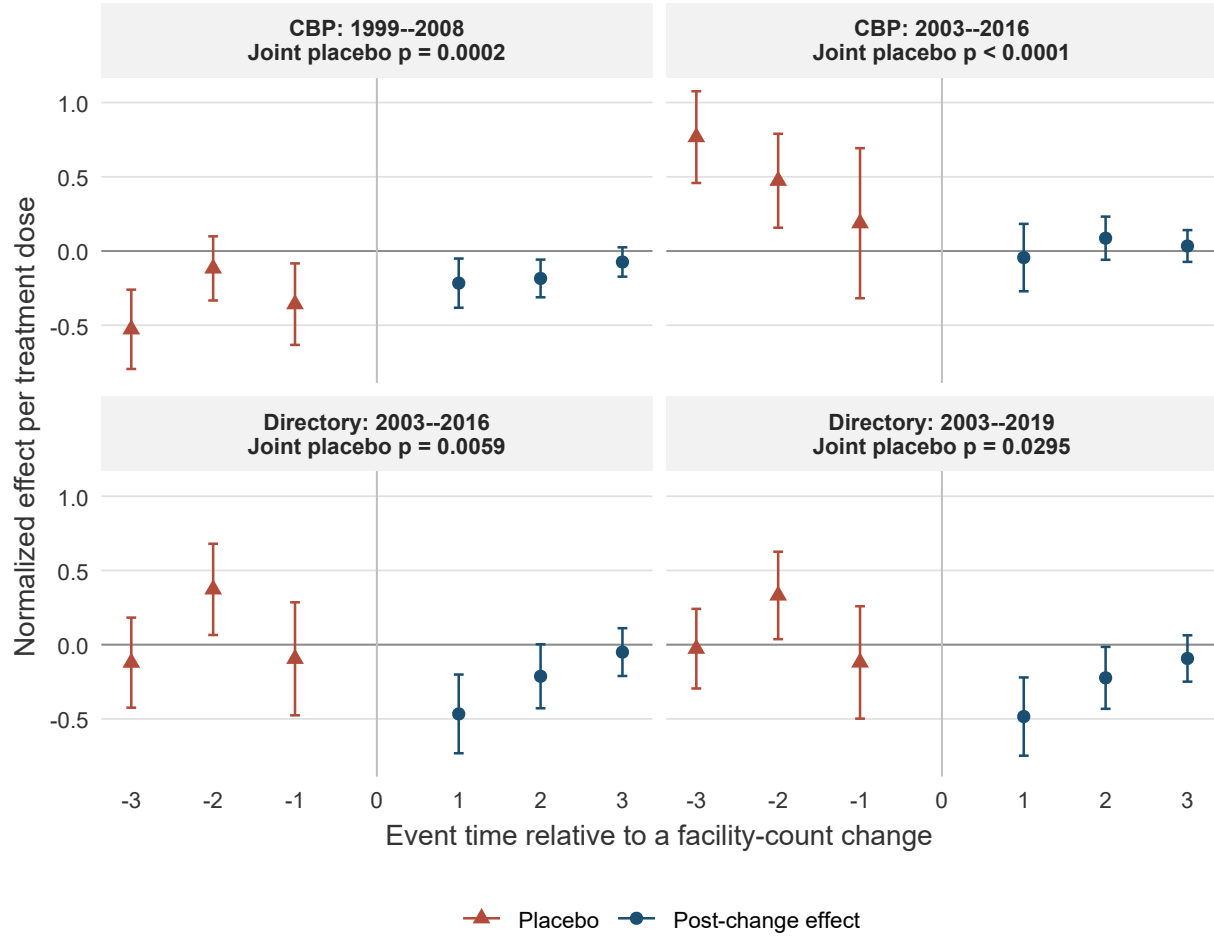
Notes: The outcome is drug-induced deaths per 100,000 residents and the exposure is the one-year-lagged facility count. The first row replicates the original 1999–2008 CBP specification. In the extension rows, the early period is 1999–2008 and the later period begins in 2009 and ends in the year shown. Directory estimates use the common CBP–directory sample through 2016; the final row extends the directory through 2019. Extension estimates come from Equation (2), with county intercepts and control coefficients allowed to differ across periods. All models include state-by-year fixed effects, use population weights, and cluster standard errors by county. The difference is the later-period coefficient minus the early-period coefficient; the final column reports the test of equality. Standard errors appear in parentheses.

Figure 1. CBP and Directory Facility Counts Agree in Levels, Not Annual Changes



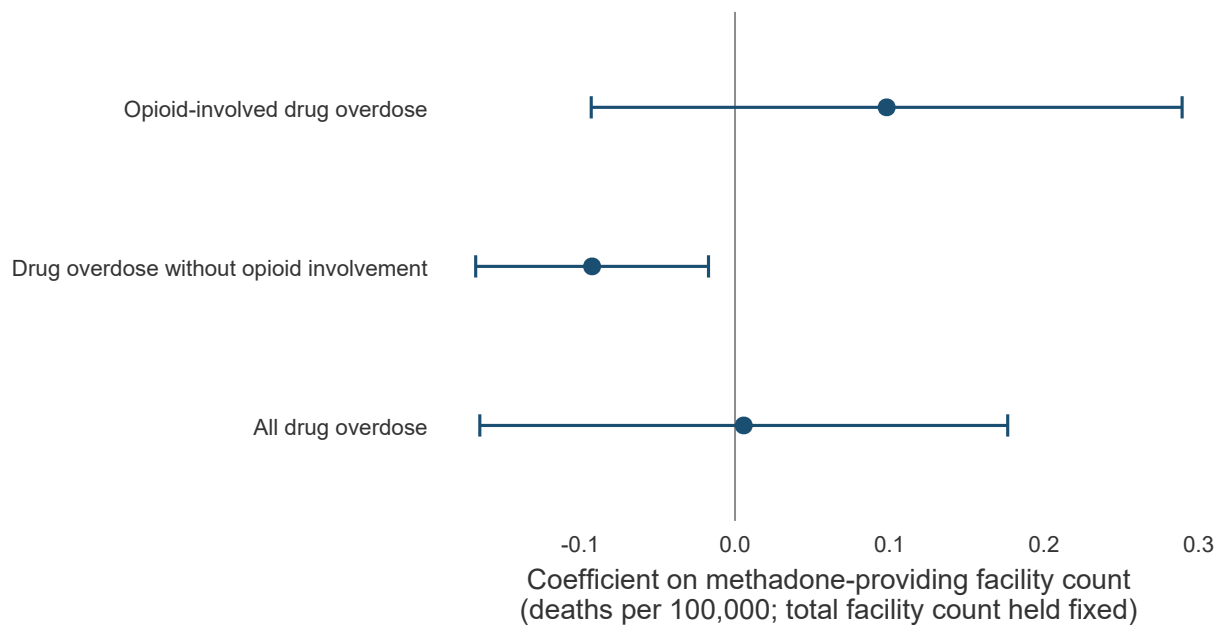
Notes: Panel A reports annual cross-sectional county correlations between the street-plus-fallback directory count and the published CBP establishment count, separately in levels and first differences. Panel B indexes the national totals in each source to one in 1998. The sample ends in 2016, before the break in broad CBP county-industry coverage. The 2001 directory is unavailable, and the plotted breaks preserve that missing survey year. Sources: U.S. Census Bureau County Business Patterns and the author's calculations from annual SAMHSA/N-SSATS substance-use treatment directories.

Figure 2. Continuous-Treatment Estimates Fail the Joint Placebo Tests



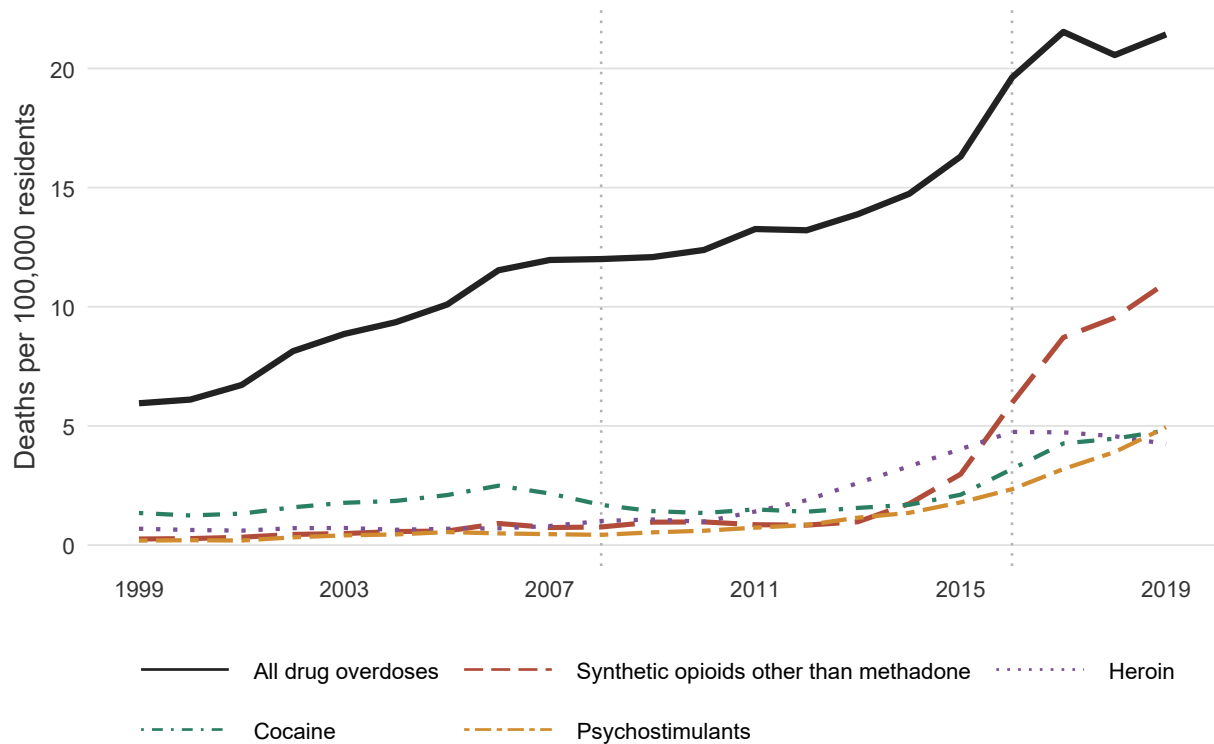
Notes: Points report normalized dynamic estimates from `did_multiplegt_dyn`; bars are 95 percent confidence intervals based on county-clustered standard errors. Negative event times are three placebo estimates before a facility-count change, and positive event times are post-change effects. Models use population weights and state-specific comparison strata. Matched samples begin in 2003 to avoid bridging the missing 2001 directory survey. Joint tests reject that the three placebo estimates are zero in all four panels. See Section 3 for the sample and estimator differences relative to Equation (1).

Figure 3. Methadone-Provision Estimates Do Not Show an Opioid-Specific Protective Association



Notes: Points report coefficients on the one-year-lagged count of methadone-providing facilities from population-weighted county-year regressions for 2003–2019; bars are 95 percent confidence intervals. Every regression also includes the total directory facility count, county fixed effects, state-by-year fixed effects, the Swensen control vector, and county-clustered standard errors. The opioid-involved and no-recorded-opioid outcomes are mutually exclusive and sum to all drug-overdose mortality. The estimates are county-level associations and should not be interpreted as clinical effects of methadone treatment.

Figure 4. The Drug Composition of Fatal Overdoses Changed Sharply



Notes: Rates equal national deaths per 100,000 residents, constructed from restricted National Center for Health Statistics multiple-cause mortality records. Drug classes are identified from contributing-cause codes and can overlap, so the class-specific series should not be added. Dashed vertical lines mark the end of Swensen's original sample in 2008 and the final year of broad CBP county-industry coverage in 2016.